# Wōdnesdæg v1.2.1

created by k.odk

# Sixty-four sines, one chord

Wōdnesdæg is sixty-four sine waves tuned to one chord and left to ring. Nothing is triggered and nothing decays: the instrument is the chord itself, and the way the waves beat against each other.

The front is two banks of sixty-four sliders. The upper bank is how loud each wave is; the lower bank is how far it sits off its own note. Push one slider off centre and you hear it beat against its neighbour; push twenty and the chord breathes.

Around the chord sit an audio file you can write on with your finger, a loop that lets the file and the drone move each other, two engines that strike the waves like pedalled strings, and a catch that throws back pieces of what was just played. Nothing in the sound uses random numbers — where something has to scatter it reads a fixed sequence, so the same settings give the same result. And nothing moves unless a switch is on.

**Too much to learn? Skip it.** Choose a key and a scale on the DRONE panel, raise a few sliders in the upper bank — not all of them — and narrow the **RANGE** bar until the low end is clean. Then take one or two sliders in the lower bank off centre and listen to the beating appear. That is the whole instrument.

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# What it needs

Sixty-four oscillators, a twelve-band filter bank, a file player and two playing engines run at once, in real time.

## System

| | |
|---|---|
| IOS | iOS 18.0 or later. |
| DEVICE | iPhone and iPad, played in landscape. On iPad the whole instrument fills the screen and every panel opens full size. |
| RECOMMENDED | iPhone 13 or newer. Wōdnesdæg is developed and tested on an iPhone 13. |
| FILES | Any audio file you can open from Files can be loaded. Recordings and saved presets are kept on the device, in the Files app under *On My iPhone › Wōdnesdæg*. |
| MICROPHONE | Only when you turn on LIVE, to play incoming sound through the instrument as it happens. Nothing is recorded or sent anywhere. Use headphones or an audio interface — a speaker feeds back into the microphone. |
| BACKGROUND | The drone keeps sounding when you switch to another app — as an instrument should. |

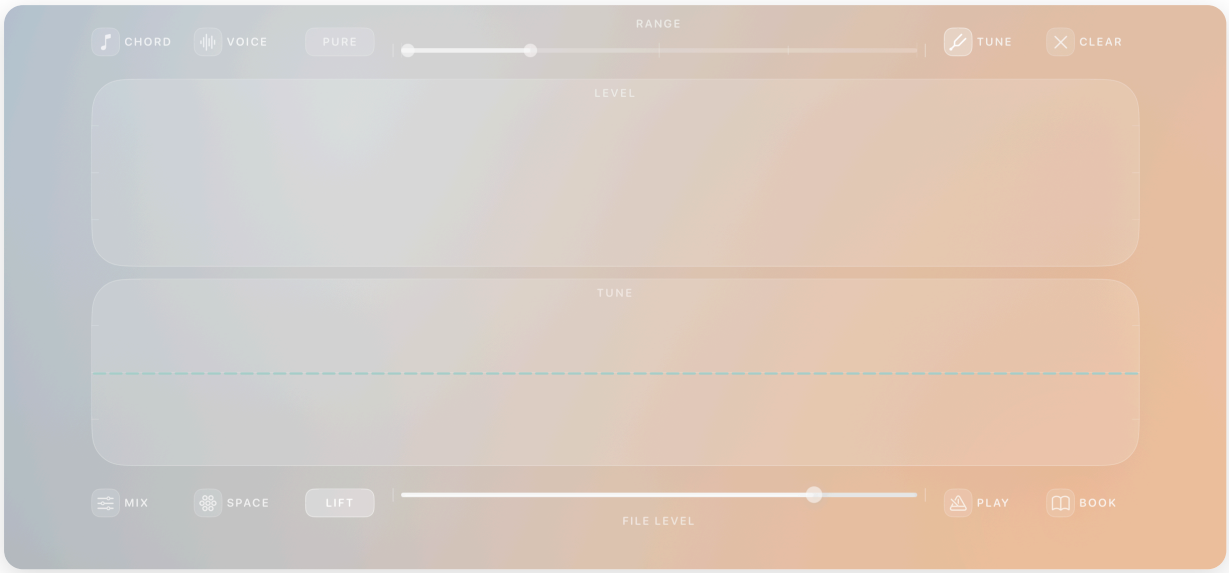
## About AUv3

Wōdnesdæg is a **standalone app**, not an AUv3 plug-in. It is meant to be one instrument you hold and play, not a sound inside a host.

## A note on listening

Sixty-four sines have nowhere to hide: small speakers will rattle on the lowest notes long before the instrument does. Headphones or proper monitors show what it really sounds like. If the low end breaks up, narrow the RANGE bar rather than turning everything down.

# Two banks and a frame of keys FRONT




### The two banks

**LEVEL** (upper) is how loud each of the sixty-four waves is — curved so the quiet end is really quiet, and weighted by pitch so a slider at the same height sounds equally loud high or low. **TUNE** (lower) is how far each wave sits off its note, centred on the dashed line, up to fifty cents either way. Draw across a bank with one finger to set many at once.

### RANGE and FILE LEVEL

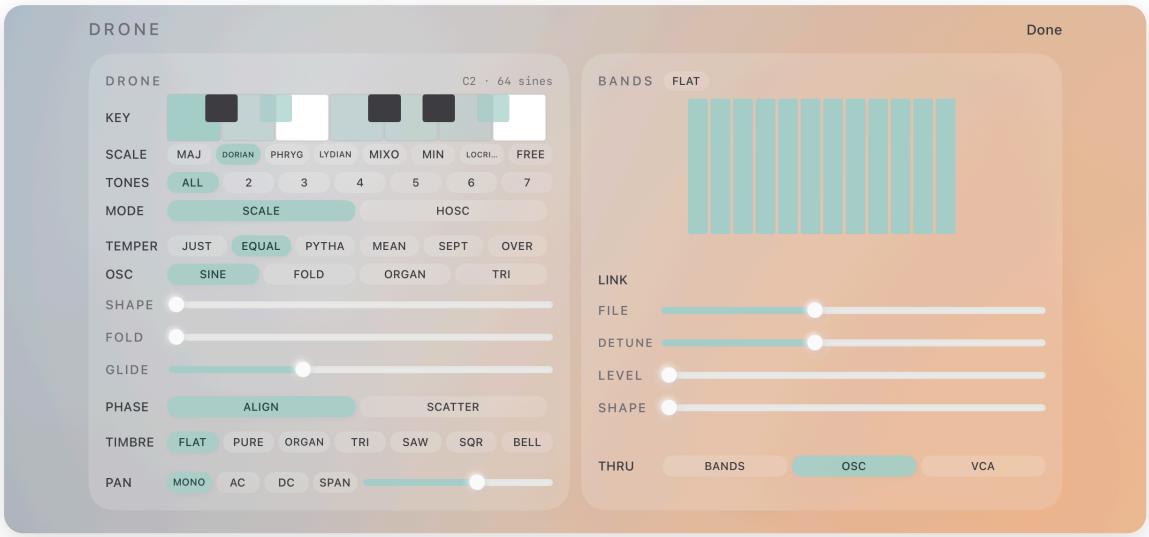
The bar on top is the range: two pointers, and the chord is only laid out between them. It starts two octaves above the root; drag the right pointer and the sines reach up into the high treble (nothing is placed above 16 kHz). The bar at the bottom is the level of the audio file.

### Keys

**CHORD** opens the DRONE panel (04). **VOICE** opens AUDIO PROCESS (05). **PURE / ANALOG** changes the waves themselves: ANALOG leans each crest slightly forward, adding a little of the second and third harmonic, like an analogue oscillator. **TUNE** steps through the temperaments; it is lit on JUST. **CLEAR** takes every slider in both banks back down.

**MIX** opens the MIXER (08). **SPACE** opens THROW (07). **ONE**: move the sliders one at a time — only the slider you first touch moves, however your finger drifts. **PLAY** opens the two playing engines (06). **BOOK** is the guide inside the app.

# Choosing the chord CHORD




**KEY** is the keyboard; the notes of the chosen scale are shaded, so you see the chord as well as the root. **SCALE** depends on the temperament; the last one, **FREE**, spreads the waves evenly over the range with no scale at all. **TONES**: how many of the scale's notes to use — with fewer, the waves double up and grow thicker, not louder.

**MODE**: SCALE deals the waves over the range like an organ is drawn — root and fifth to the top, sevenths and seconds left lower down. HOSC is eight harmonic oscillators, each on one note of the scale carrying its own first to eighth harmonic; it stays low and thick.

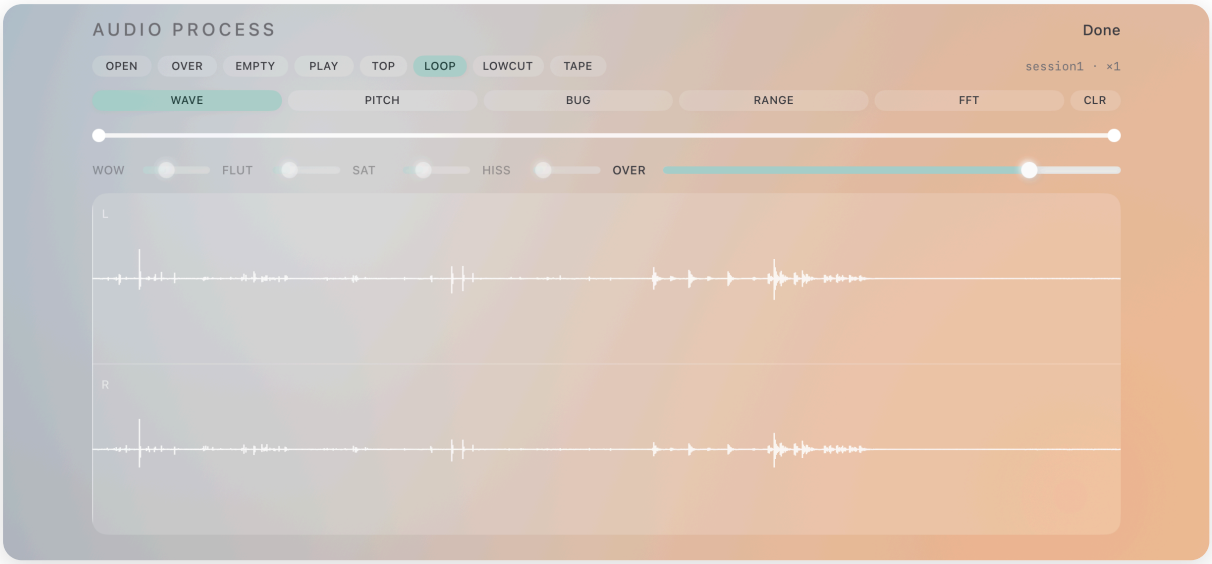
**TEMPER**: **JUST** (pure thirds), **EQUAL**, **PYTHA** (stacked fifths — the thirds grind), **MEAN** (quarter-comma), **SEPT** (seven-limit), **OVER** (the harmonic series itself).

**OSC** is the shape of one wave — SINE, FOLD, ORGAN, TRI — and **SHAPE** bends between the sine and it. **FOLD** folds the crest back into itself. **GLIDE**: how long a wave takes to reach a new pitch. **PHASE**: ALIGN starts all sixty-four together (a stronger low end), SCATTER fans them out. **TIMBRE** chooses which harmonics of the chord get through. **PAN** places the sixty-four — MONO, AC, DC, SPAN — and the slider says how wide.

### BANDS and LINK

Twelve band-pass filters, 60 Hz to 6.5 kHz, on the drone; **FLAT** opens them all. **LINK** lets the file and the drone move each other: **FILE** (the drone's beating moves the file's speed), **DETUNE** (the file's level detunes the waves), **LEVEL** (the file's spectrum comes back to the drone) and **SHAPE**. **LIFT**, beside the LINK title: while the file plays, waves near its loudest bands rise even if their sliders are down. **THRU** says where LEVEL lands: the BANDS, the OSC (lifting waves), or a **VCA** that opens only when the file — or LIVE input — sounds: a gate that ignores the room's noise floor and opens clearly when sound arrives; LEVEL sets how far it closes.

# Load it, loop it, write on it VOICE




### The top row

**OPEN** reads a file. **OVER** reads another file on top of it — the higher the OVER slider, the less it is a mix: the sum is folded, the two are ring-modulated, and at the top their sixteen-bit numbers are crossed together. **EMPTY** throws the file away. **PLAY**, **TOP** (back to the loop start), **LOOP**, **LOWCUT** (takes the bottom off, where the drone lives), **TAPE**, and **LIVE** — incoming sound from the mic or an interface flows through in real time, in place of the file, through the same LOWCUT, tape, bands and LINK. Nothing is recorded. The bar beneath is the loop: only what lies between the two pointers plays.

### TAPE

**WOW** slow and **FLUT** fast wobble in the timing, left and right separately; **SAT** squashes and takes the top off; **HISS** is the rub.

### Writing on the sound

The chips choose what the band shows and what your finger writes. **WAVE**: the sound itself, left over right — the half you start in is the channel you write. The line from the last point to this one is filled in, and the ends fade into what was there. **PITCH**: how much the reading speed is pushed, up to four octaves. **BUG**: where the reading breaks — jumps, runs backwards, holds, inverts, crushes. **RANGE**: drag to set the loop. **FFT**: draw a shelf from 20 Hz to 20 kHz. **CLR** clears whichever is showing. Pinch to zoom; close in, the samples themselves are drawn.

# Two engines, A and B PLAY




Instead of holding all sixty-four on, PLAY treats them as strings and strikes them. A struck string rises over **ATTACK**, then is left to **RELEASE** — up to half a minute — so several chords lie over each other, like a piano with the pedal down. The quietest string is always the one struck next, and only waves whose LEVEL slider is up are ever played. Between strikes the chord moves to another degree of the scale. A and B each have a switch, a mode and an envelope of their own.

### MODE

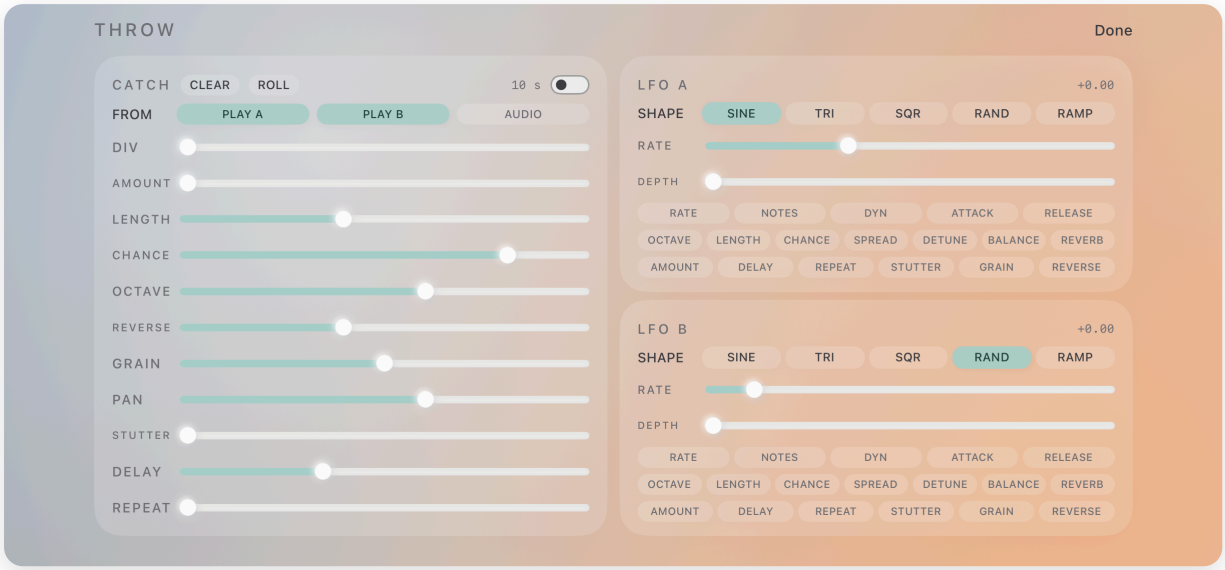
**CLOCK**: its own clock; **SWING** pushes every other strike. **DRIFT**: the gap wanders; **LOCK** closes the wandering into a loop of sixteen, then eight, four. **EUCLID**: sixteen steps with **FILL** strikes spread evenly. **TRIG**: struck by the audio file itself — **SENS** is how small a rise counts, **HOLD** the shortest gap. **SPRAY**: mostly quiet, then a handful at once (**BURST**).

### The rows

**RATE**: the gap, an eighth of a second to six seconds. The second row's name follows the mode, so nothing moves under your finger when you switch. **NOTES**: one to five strings each time. **ROLL**: at zero they land together, up and it becomes an arpeggio. **MOTION**: how far the chord may move — low, and it circles the tonic and the fifth. **DYN**: how much the strikes vary in strength.

**ATTACK**: three milliseconds to two seconds — long, and a strike becomes a swell. **RELEASE**: six tenths of a second to thirty; this decides how much overlaps.

# The last ten seconds, thrown back SPACE




## CATCH

The last ten seconds of the dry sound are always held. Each time the playing engines strike, a piece of it is cut and thrown back — at another octave, backwards, or scattered as grains. Only the dry sound is held, so a throw never throws itself back. **FROM** chooses what gives the signal: PLAY A, PLAY B, or AUDIO (the file rising). **DIV**: how many signals per throw.

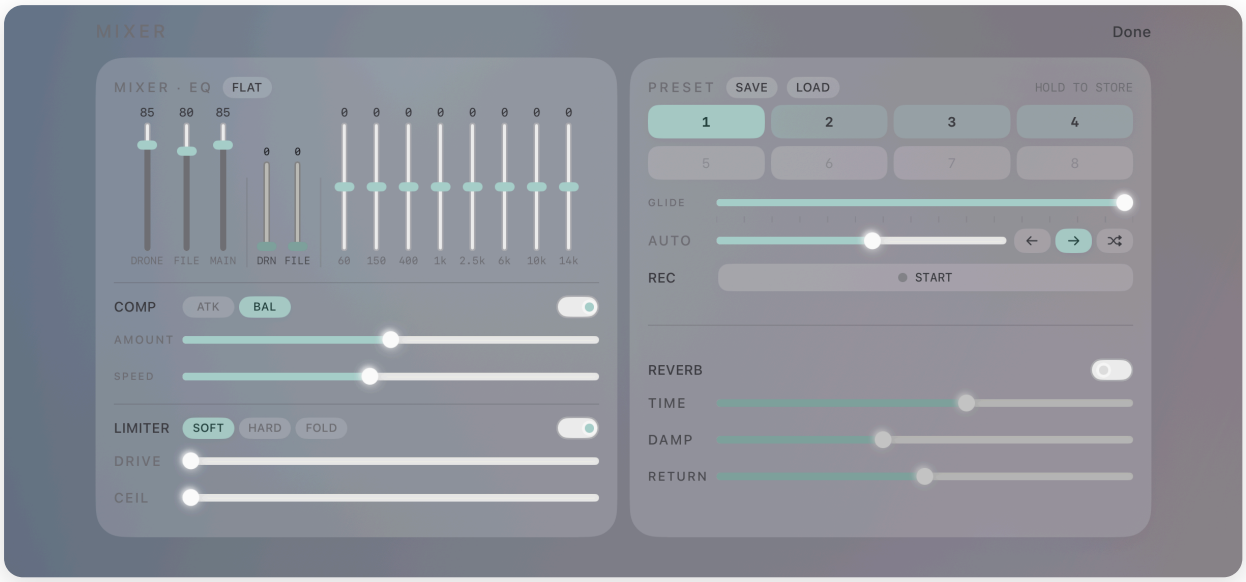
**AMOUNT** is how much is heard. **LENGTH** of a piece, **CHANCE** that one comes at all, **OCTAVE** how far it may be moved, **REVERSE** how often backwards, **GRAIN** how often as grains, **PAN** how wide, **STUTTER** how often it breaks, **DELAY** and **REPEAT** to throw a piece again. **CLEAR** empties the ten seconds; **ROLL** redraws every knob but AMOUNT; the switch does it on every throw.

## LFO A · LFO B

Two slow waves — **SINE**, **TRI**, **SQR**, **RAND** (a stepped fixed sequence, not dice) and **RAMP** — with **RATE** and **DEPTH**. The chips below are where each goes, as many as you like.

The playing knobs — **RATE**, **NOTES**, **DYN**, **ATTACK**, **RELEASE** — cycle A → B → AB → off with each tap, so you choose which engine is moved. The others are on or off: **OCTAVE**, **LENGTH**, **CHANCE**, **SPREAD**, **DETUNE**, **BALANCE**, **REVERB** and the throw's own **AMOUNT**, **DELAY**, **REPEAT**, **STUTTER**, **GRAIN**, **REVERSE**. The number top right shows where each wave is now.

# The desk, levelling, and a room MIX




### MIXER · EQ

One row: **DRONE**, **FILE** and **MAIN** (FILE only sets how loud the file or LIVE input is heard; LINK, the VCA and TRIG still read it at full level, so you can gate the drone with a voice you do not hear); the two sends to the reverb, **DRN** and **FILE** (the drone and the file are sent separately, after their faders); and an eight-band **EQ** on the file, 60 Hz to 14 kHz, ±12 dB, gentle and overlapping like a tone control. **FLAT** resets it.

### COMP · LIMITER

A drone has no rests, so the way it is levelled matters. **COMP**: **ATK** follows the attack, **BAL** rides the average; **AMOUNT** and **SPEED**. The **LIMITER** is last before the output and on by default — **SOFT** rounds the top, **HARD** stops it, **FOLD** turns it back and is a sound rather than a safeguard. **DRIVE** pushes into it; **CEIL** is where the ceiling sits.

### REVERB

A plain reverb, four combs and two all-passes each side. It is not in the signal path: only what the sends feed goes in. **TIME** is how long it rings, **DAMP** how quickly the top is lost, **RETURN** how much comes back to the desk. Switching it off empties the tail rather than letting it hang.

### PRESET

The other half of this card is section 09.

# Keeping it, and letting it move

### Eight slots

On the MIXER's PRESET card. **Hold** a number to store what is sounding now; a slot that holds something is tinted pale mint with a bold number; the one you are on is full mint. **Tap** to go there. A slot keeps everything: both banks, key, scale and temperament, RANGE, the DRONE, PLAY, THROW and LFO settings, the file's knobs and the lines you drew, the bands, the desk, EQ, COMP, LIMITER and REVERB. Only the audio file itself and the zoom are not kept.

### GLIDE

How long the move takes — instant at the left, a full minute at the right. The knobs and sliders slide over that time, and the two banks on the front drift across as a soft ghost until they settle on the new values. Waves whose pitch changes do not slide in frequency: each fades out on its old note and back in on its new one, one after another, so the chord changes the way a modulation does — notes common to both stay sounding throughout. The ticks under GLIDE fill in as the move goes. You can touch any slider while it moves — whatever you touch stops there and stays where you put it. Tap another slot mid-move and it carries on from the sound as it is.

### AUTO

Moves to another slot by itself, every three minutes (left) to ten (right). ← goes to the previous slot, → to the next, the shuffle to any other — empty slots are skipped. Tap the lit button again to stop. It keeps going in the background.

### SAVE and LOAD

**SAVE** beside the title stores all eight slots, with GLIDE and AUTO, as one named set on this iPhone; a name that exists is never overwritten. **LOAD** lists the sets, newest first — tap one and the eight slots are replaced; the sound does not change until you tap a slot.

### REC

Records what comes out of the desk as a 24-bit WAV. Tap to start; it shows the time. Tap again to stop, and a share sheet opens to save or send the file. Every recording also stays in *Files › On My iPhone › Wōdnesdæg › Recordings*.

# Version 1.2.1

CURRENT

## 1.2.1

iPhone and iPad, landscape · iOS 18.0 or later. This guide describes this version.

### What’s new in 1.2.1

| | |
|---|---|
| **LIVE** | Incoming sound from the mic or an interface flows through the file's path in real time. With THRU on VCA it gates the drone clearly; FILE only sets how loud it is heard. |
| **Presets** | Smoother moves: no dips or clicks; waveform, ANALOG, pan and COMP/LIMITER blend too; a new slot mid-move carries on from the sound as it is; sliders can be touched during a move. |
| **ONE** | Moves the front sliders one at a time. LIFT has moved to LINK on the DRONE panel. |
| **Front** | Clearer labels and stored slots; more room above and below on iPad. |

### 1.2.0

| | |
|---|---|
| **iPad** | Wōdnesdæg runs on iPad, in landscape and full screen; every panel opens with the iPhone layout, scaled up. |
| **Presets** | Eight slots that keep everything, with GLIDE up to a minute and chord changes that move like a modulation. |
| **AUTO** | Moves between slots by itself every three to ten minutes — left, right or shuffled. |
| **SAVE · LOAD** | Named sets of all eight slots, kept on the device. |
| **REC** | Records the output as a 24-bit WAV. |
| **MIXER** | The desk and the EQ share one card; REVERB is down to TIME, DAMP and RETURN. |
| **RANGE** | Reaches up to 16 kHz; it still starts at two octaves. |
| **Sound** | The output's soft clipping only acts near the top, so sixty-four sines no longer buzz against each other. |

Written in Swift with AVAudioEngine, with no third-party audio code. The band bank and the harmonic oscillator mode are written from how such circuits behave; no maker’s name or model appears in the app. Wōdnesdæg, Old English for Wednesday.

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WŌDNESDÆG

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